Unbeatable Lesson 1

Introduction to Statistics and Basketball Data

Lesson objectives:
• Understand basic statistical concepts: mean, median, and mode • Calculate mean, median, and mode • Analyze basketball player statistics
Assessment:
• Handout • Exit Ticket
Key Points:
• Introduce Unbeatable Initiative • Review history of basketball • Introduce key terms (mean, median, mode). • Analyze a set of basketball player statistics (points scored, assists, rebounds). • Group activity: Calculate the mean, median, and mode of selected statistics. • Importance of statistics in analyzing player performance.

Component:	Teacher & Student Actions	Materials
Do Now	• Question: Do you have a favorite basketball player? Who is it? What factors do you think contribute to a player's success in a game?	Piece of paper, journal or Handouts
Introduction of Unbeatable	• Introduce students to the initiative and the national competition	Slides Note to Students
Mini-Lesson	• Define mean, median and mode and provide sample examples • Relate the concepts to their applicability in basketball	Slides Handouts
Guided Practice	• Using a set of data (players stats in shots, assists, and rebounds) guide students through finding the mean median and mode of shots.	Slides Handouts

Independent Practice	• Working independently or in groups, instruct students to solve for the mean median and mode of data for assists and rebounds.	Slides Handouts
Closing	• Exit Ticket: Why might statistics be important in basketball?	Piece of paper, journal or Handouts

Differentiation Considerations:

- Choose either paper/pen/pencil or digital tools (Google Slides, Google Draw, etc)
- Students can work individually or in groups
- Students can have more teacher support where needed

Standard(s):

Common Core Standards Math

- **CCSS.MATH.CONTENT.6.SP.A.2** Understand that a set of data collected to answer a statistical question has a distribution which can be described by its center, spread, and overall shape.
- **CCSS.MATH.CONTENT.6.SP.A.3**
- Recognize that a measure of center for a numerical data set summarizes all of its values with a single number, while a measure of variation describes how its values vary with a single number.
- **CCSS.MATH.CONTENT.6.SP.B.5.C**
- Summarize numerical data sets in relation to their context, such as by: Giving quantitative measures of center (median and/or mean) and variability (interquartile range and/or mean absolute deviation), as well as describing any overall pattern and any striking deviations from the overall pattern with reference to the context in which the data were gathered.

Common Core Standards ELA

- **CCSS.ELA-LITERACY.W.6-12.10** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.
- **CCSS.ELA-LITERACY.L.6-8.6**
- Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression.
- **CCSS.ELA-LITERACY.L.9-12.6**
- Acquire and use accurate general academic and domain-specific words and phrases, sufficient for reading, writing, speaking, and listening at the college and career readiness level; demonstrate independence in gathering vocabulary knowledge when considering a word or phrase important to comprehension or expression.

Unbeatable Lesson 2
Understanding Probability

Lesson objectives:
• Explain the concept of probability. • Apply probability to basketball scenarios, for example team win percentages and predicting game outcomes • Understand how win percentages relate to March Madness brackets
Assessment:
• Handout • Exit Ticket
Key Points:
• Definition of probability and examples in basketball • Discussion of how probability affects predicting outcomes, game decisions, and player behavior. • Calculating probability using real-life basketball data of winning percentage, shooting probability. • Introduction to the March Madness competition.

Component:	Teacher & Student Actions	Materials
Do Now	• Question: What 5 words or phrases come to mind when you read "March Madness"?	Piece of paper, journal or Handouts
Mini-Lesson & Guided Practice	• Define probability, win percentage, outcomes, and brackets • Introduce probability and carry-out playing deck probability demonstration • Discuss how probability plays a role in basketball outcomes, real-life situations, and decision-making • Introduce concept of winning percentage	Slides Handouts
Independent Practice	• March Madness metrics activity: Students will use provided data to analyze five NCAA teams of their choice. They will then choose one of their teams to square off in a "game" of best winning percentage. ◦ Note: All data is provided within the handout. Optional revision to use current statistics for the	Slides Handouts

	24-25 season (available online at ESPN.com)	
Closing	• Exit Ticket: How do team win percentages affect predictions in March Madness?	Piece of paper, journal or [Handouts]

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Standard(s):

Common Core Standards Math

- **CCSS.MATH.CONTENT.7.SP.C.5** Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around 1/2 indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.
- **CCSS.MATH.CONTENT.7.SP.C.6** Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability.
- **CCSS.MATH.CONTENT.7.SP.C.7** Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy.
- **CCSS.MATH.CONTENT.7.SP.C.8** Find probabilities of compound events using organized lists, tables, tree diagrams, and simulation.
- **CCSS.MATH.CONTENT.6.SP.B.5.A** Summarize numerical data sets in relation to their context, such as by: Reporting the number of observations.

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Unbeatable Lesson 3

Calculating Shooting Percentages & Point Differential

Lesson objectives:		
• Calculate shooting percentages • Calculate point differential • Understand the significance of shooting percentages and point differentials in basketball		
Assessment:		
• Handout • Exit Ticket		
Key Points:		
• Introduce key terms: shooting percentage and point differential • Guide students through practice of calculating shooting percentage and point differential • Carryout trashketball activity (or basketball game) and discuss statistics of game		
Component:	Teacher & Student Actions	Materials
Do Now	• Question: Who is your favorite basketball player? Why? What skills are they particularly good at?	Piece of paper, journal or Handouts
Mini-Lesson & Guided Practice	• Define shooting percentage and point differential • Guide students through problems solving for shooting percentage and point differential	Slides Handouts
Independent/Group Practice	• Trashketball Activity: Divide students into groups or have them play individually. Each “team” (groups or individuals) will take 10 turns attempting to make a shot into a trashcan/at a target on the board/or similar. Students will record the data for each team and complete their activity handout. • Discussion about trashketball • SPECIAL NOTE: If possible, and you have access to a basketball court, consider hosting a real basketball game. There can be players and stat-keepers.	Slides Handouts

Closing	• Exit Ticket: If you were to change one rule of the Trashketball game (e.g., distance from the basket, number of attempts), how might that affect the probabilities and outcomes? Explain your reasoning.	Piece of paper, journal or Handouts

Differentiation Considerations:
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Standard(s):

Common Core Standards Math

- **CCSS.MATH.CONTENT.6.SP.B.5.A** Summarize numerical data sets in relation to their context, such as by: Reporting the number of observations.
- **CCSS.MATH.CONTENT.6.SP.B.5.B** Describing the nature of the attribute under investigation, including how it was measured and its units of measurement.
- **CCSS.MATH.CONTENT.7.SP.A.1** Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences.
- **CCSS.MATH.CONTENT.7.SP.A.2** Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions.

Common Core Standards ELA

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Unbeatable Lesson 4
Visualizing Data & Analyzing Player Statistics

Lesson objectives:		
• Analyze player statistics using visual representations such as line graphs and bar graphs. • Identify trends in player performance over time to make informed predictions • Discuss the importance of visualizing data in sports analysis		
Assessment:		
• Handout • Exit Ticket		
Key Points:		
• Introduce key terms bar graph, line graph, and trend • Guide students through practicing making graphs • Student activity of using provided data to create a graph of the data of their choice		
Component:	Teacher & Student Actions	Materials
Do Now	• Question: What makes a player valuable to a team?	Piece of paper, journal or Handouts
Introduction of Unbeatable	• Introduce students to the initiative and the national competition	Slides Note to Students
Mini-Lesson & Guided Practice	• Define bar graph, line graph, and trend • Connect how these graphs can be used to represent basketball statistics/data • Guide students through creating graphs using the handout	Slides Handouts
Independent Practice	• Working independently or in groups, instruct students to use the provided data to create their own graphs. Students may choose to create a bar graph or a line graph, using any player statistic. Offer suggestions and assist students in selecting what data to graph. • If time allows, lead a reflective discussion about students' graphs.	Slides Handouts

Closing	• Exit Ticket: Why might visual representations be important and useful in math?	Piece of paper, journal or Handouts

Differentiation Considerations:

- Choose either paper/pen/pencil or digital tools (Google Slides, Google Draw, etc)
- Students can work individually or in groups
- Students can have more teacher support where needed: Assist individual students with additional guidance on selecting a statistic to graph

Standard(s):

Common Core Standards Math

- **CCSS.MATH.CONTENT.7.SP.A.1** Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences.
- **CCSS.MATH.CONTENT.7.SP.B.3** Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability.
- **CCSS.MATH.CONTENT.8.SP.A.2** Know that straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line.
- **CCSS.MATH.CONTENT.8.SP.A.4** Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table. Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. Use relative frequencies calculated for rows or columns to describe possible association between the two variables.

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Unbeatable Lesson 5
Dream Team Assembly

Lesson objectives:
• Synthesize knowledge of statistics to create and present a Dream Team • Justify why they chose the players they did • Use data to inform their Dream Team assembly • Engage in peer feedback and constructive criticism
Assessment:
• Handout • Exit Ticket
Key Points:
• Students will curate their Dream Team, justify their choices and design team jerseys. All students can submit their Dream Teams/ jerseys to the BreakFree contest.

Component:	Teacher & Student Actions	Materials
Do Now	• Question: What statistic do you believe is the most important for evaluating a player's overall performance? Why?	Piece of paper, journal or Handouts
Introduction to Unbeatable Final Project Competition	• Introduce students to the final product of creating their Dream Team starting five, writing an explanation for the team, and designing a team jersey • Review/Skim the slides with player data • Distribute handout with player data	Slides Note to Students Player Data Handout
Independent Practice	• Students will conduct research (using provided materials or online resources if available) and analysis to determine their dream team and complete the final submission worksheet (in handouts and linked contest policies) • Be sure to review contest rubric and policies • Allow time for students to present their dream teams • SPECIAL NOTE: Player data sheets will be added to slides and materials by February	Slides Handouts Contest Policies and Guidelines

	15, 2025. This is to allow for statistics for the 24-25 NBA and WNBA seasons to be collected. ○ If you have downloaded or accessed these slides prior to this (and are seeing this note) please check the Unbeatable website for updated materials! www.breakfree-ed.org/unbeatable	
Closing	• Exit Ticket: What criteria did you use to select your players? Are there any surprises in your choices?	Piece of paper, journal or Handouts

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Common Core Standards ELA

- **CCSS.ELA-LITERACY.W.6-12.2** Write informative/explanatory texts to examine a topic and convey ideas, concepts, and information through the selection, organization, and analysis of relevant content.
- **CCSS.ELA-LITERACY.W.6-12.4**
- Produce clear and coherent writing in which the development, organization, and style are appropriate to task, purpose, and audience.
- **CCSS.ELA-LITERACY.W.6-12.10** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.